Anna Ma

Associate Professor
Department of Mathematics
University of California, Irvine
Email: anna.ma@uci.edu
Website: anna-math.github.io

My research focuses on the problems that arise in mathematical data science. I am particularly interested in designing and analyzing iterative algorithms for large-scale data, as well as leveraging and developing tools in numerical linear algebra, signal processing, machine learning, and probability.

Academic Positions

Associate Professor — July 2026 - Current
Department of Mathematics
University of California, Irvine

Assistant Professor — July 2022 - June 2026
Department of Mathematics
University of California, Irvine

Visiting Assistant Professor — July 2019 - June 2022
Department of Mathematics
University of California, Irvine
Faculty Mentor: Roman Vershynin

UC Chancellor's Postdoctoral Fellow — July 2018 - June 2019
Department of Mathematics,
University of California, San Diego
Faculty Mentor: Rayan Saab

Education

PhD **Claremont Graduate University**, Computational Science — Sept. 2013 – June 2018
Jointly with San Diego State University
Dissertation: "Stochastic Iterative Algorithms for Large-Scale Data"
Thesis Adviser: Deanna Needell

BS **University of California, Los Angeles**, Mathematics — Sept. 2009 – June 2013
Minor in Statistics
Specialization in Computing

Publications

Accepted Publications

[1] A. Ma, A. Flenner, D. Needell, and A. G. Percus. "Improving image clustering using sparse text and the wisdom of the crowds". *2014 48th Asilomar Conference on Signals, Systems and Computers*. IEEE. 2014, pp. 1555–1557. DOI: 10.1109/ACSSC.2014.7094725.

[2] A. Ma, D. Needell, and A. Ramdas. "Convergence properties of the randomized extended Gauss–Seidel and Kaczmarz methods". *SIAM Journal on Matrix Analysis and Applications* 36.4 (2015), pp. 1590–1604. DOI: 10.1137/15M1014425.

[3] D. Baron, A. Ma, D. Needell, C. Rush, and T. Woolf. "Conditional approximate message passing with side information". *2017 51st Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2017, pp. 430–434. DOI: 10.1109/ACSSC.2017.8335374.

[4] A. Ma and D. Needell. "A gradient descent approach for incomplete linear systems". *2018 52nd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2018, pp. 764–768. DOI: 10.1109/ACSSC.2018.8645206.

[5] A. Ma, D. Needell, and A. Ramdas. "Iterative methods for solving factorized linear systems". *SIAM Journal on Matrix Analysis and Applications* 39.1 (2018), pp. 104–122. DOI: 10.1137/17M1115678.

[6] S. Ravishankar, A. Ma, and D. Needell. "Analysis of Fast Alternating Minimization for Structured Dictionary Learning". *2018 Information Theory and Applications Workshop (ITA)*. IEEE. 2018, pp. 1–9. DOI: 10.1109/ITA.2018.8503246.

[7] Y. Van Gennip, B. Hunter, A. Ma, D. Moyer, R. De Vera, and A. L. Bertozzi. "Unsupervised record matching with noisy and incomplete data". *International Journal of Data Science and Analytics* 6.2 (2018), pp. 109–129. DOI: 10.1007/s41060-018-0129-7.

[8] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. "Compressed Anomaly Detection with Multiple Mixed Observations". *Research in Data Science*. Springer, 2019, pp. 211–237. DOI: 10.1007/978-3-030-11566-1_10.

[9] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. "Fast hyperspectral diffuse optical imaging method with joint sparsity". *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. 2019, pp. 4758–4761. DOI: 10.1109/EMBC.2019.8857069.

[10] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. "Jointly Sparse Signal Recovery with Prior Info". *2019 53rd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2019, pp. 645–649. DOI: 10.1109/IEEECONF44664.2019.9048818.

[11] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. "Sparse randomized Kaczmarz for support recovery of jointly sparse corrupted multiple measurement vectors". *Research in Data Science*. Springer, 2019, pp. 1–14. DOI: 10.1007/978-3-030-11566-1_1.

[12] A. Ma, Y. Zhou, C. Rush, D. Baron, and D. Needell. "An approximate message passing framework for side information". *IEEE Transactions on Signal Processing* 67.7 (2019), pp. 1875–1888. DOI: 10.1109/TSP.2019.2899286.

[13] D. Needell and A. Ma. "Stochastic Gradient Descent for Linear Systems with Missing Data". *Numerical Mathematics: Theory, Methods and Applications* 12.1 (2019). DOI: 10.4208/nmtma.OA-2018-0066.

[14] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. "Stochastic iterative hard thresholding for low-Tucker-rank tensor recovery". *2020 Information Theory and Applications Workshop (ITA)*. IEEE. 2020, pp. 1–5. DOI: 10.1109/ITA50056.2020.9244965.

[15] S. Ravishankar, A. Ma, and D. Needell. "Analysis of fast structured dictionary learning". *Information and Inference: A Journal of the IMA* 9.4 (2020), pp. 785–811. DOI: 10.1093/imaiai/iaz028.

[16] J. A. De Loera, J. Haddock, A. Ma, and D. Needell. "Data-driven algorithm selection and tuning in optimization and signal processing". *Annals of Mathematics and Artificial Intelligence* 89.7 (2021), pp. 711–735. DOI: 10.1007/s10472-020-09717-z.

[17] N. Durgin, R. Grotheer, C. Huang, S. Li, A. Ma, D. Needell, and J. Qin. "A Simple Recovery Framework for Signals with Time-Varying Sparse Support". *Advances in Data Science*. Springer, 2021, pp. 211–230. DOI: 10.1007/978-3-030-79891-8_9.

[18] J. Haddock and A. Ma. "Greed works: An improved analysis of sampling Kaczmarz–Motzkin". *SIAM Journal on Mathematics of Data Science* 3.1 (2021), pp. 342–368. DOI: 10.1137/19M1307044.

[19] A. Ma and D. Molitor. "Randomized Kaczmarz for tensor linear systems". *BIT Numerical Mathematics* (2021), pp. 1–24. DOI: 10.1007/s10543-021-00877-w.

[20] J. Qin, S. Li, D. Needell, A. Ma, R. Grotheer, C. Huang, and N. Durgin. "Stochastic greedy algorithms for multiple measurement vectors". *Inverse Problems & Imaging* 15.1 (2021), p. 79. DOI: 10.3934/ipi.2020066.

[21] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. "Iterative hard thresholding for low cp-rank tensor models". *Linear and Multilinear Algebra* 70.22 (2022), pp. 7452–7468. DOI: 10.1080/03081087.2021.1992335.

[22] E. Lybrand, A. Ma, and R. Saab. "On the number of faces and radii of cells induced by Gaussian spherical tessellations". *Applied and Computational Harmonic Analysis* 56 (2022), pp. 176–188. DOI: 10.1016/j.acha.2021.08.005.

[23] K. Dover, Z. Cang, A. Ma, Q. Nie, and R. Vershynin. "AVIDA: An alternating method for visualizing and integrating data". *Journal of Computational Science* 68 (2023), p. 101998. DOI: 10.1016/j.jocs.2023.101998.

[24] R. Grotheer, S. Li, A. Ma, D. Needel, and J. Qin. "Stochastic natural thresholding algorithms". *2023 57th Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2023, pp. 832–836. DOI: 10.1109/IEEECONF59524.2023.10477079.

[25] J. Haddock, A. Ma, and E. Rebrova. "On subsampled quantile randomized Kaczmarz". *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–8. DOI: 10.1109/Allerton58177.2023.10313381.

[26] C. Huynh, A. Ma, and M. Strand. "Block-missing data in linear systems: An unbiased stochastic gradient descent approach". *2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*. IEEE. 2023, pp. 1–7. DOI: 10.1109/Allerton58177.2023.10313429.

[27] A. Ma, D. Stöger, and Y. Zhu. "Robust recovery of low-rank matrices and low-tubal-rank tensors from noisy sketches". *SIAM Journal on Matrix Analysis and Applications* 44.4 (2023), pp. 1566–1588. DOI: 10.1137/22M150071X.

[28] E. H. Bergou, S. Boucherouite, A. Dutta, X. Li, and A. Ma. "A Note on the Randomized Kaczmarz Algorithm for Solving Doubly Noisy Linear Systems". *SIAM Journal on Matrix Analysis and Applications* 45.2 (2024), pp. 992–1006. DOI: 10.1137/23M155712X.

[29] R. Grotheer, S. Li, A. Ma, D. Needell, and J. Qin. "Iterative singular thresholding algorithms for tensor recovery". *Inverse Problems and Imaging* 18.4 (2024), pp. 889–907. DOI: 10.3934/ipi.2023059.

[30] E. Battaglia and A. Ma. "Quantile Randomized Kaczmarz with Time-Varying Noise in Signal". *To appear in Proceedings of the IEEE Asilomar Conference on Signals, Systems and Computers*. 2025.

[31] E. Battaglia and A. Ma. "Reverse Quantile-RK and Its Application to Quantile-RK". *Numerical Linear Algebra with Applications* 32.3 (2025), e70024. DOI: https://doi.org/10.1002/nla.70024.

[32] H. Luo and A. Ma. "Frontal Slice Approaches for Tensor Linear Systems". *Numerical Mathematics: Theory, Methods and Applications* 18.2 (2025), pp. 353–394. DOI: 10.4208/nmtma.OA-2024-0096.

[33] A. Ma, D. Needell, and A. Xue. "Stochastic Gradient Descent on Tensors with Missing Data". *To appear in Proceedings of the AAAI 2025 Workshop on CoLoRAI*. 2025.

[34] E. Battaglia and A. Ma. "Quantile-RK and double quantile-RK error horizon analysis". *Linear Algebra and its Applications* 736 (2026), pp. 284–308. DOI: 10.1016/j.laa.2026.01.032.

[35] J.-F. Cai, J. Chen, A. Ma, and T. Wu. "On the Subsample Size of Quantile-Based Randomized Kaczmarz". *SIAM Journal on Matrix Analysis and Applications* 47.2 (2026), pp. 802–823. DOI: 10.1137/25M1785678.

[36] H. Luo, A. Ma, L. Stephan, and Y. Zhu. "Wedge Sampling: Efficient Tensor Completion with Nearly-Linear Sample Complexity". *Proceedings of Thirty-Ninth Conference on Learning Theory*. Ed. by S. Hanneke and T. Lattimore. Vol. 336. Proceedings of Machine Learning Research. PMLR, 29 Jun–03 Jul 2026, pp. 4883–4884. URL: https://proceedings.mlr.press/v336/luo26a.html.

[37] A. Ma, D. Needell, and A. Xue. "Stochastic Gradient Descent for Incomplete Tensor Linear Systems". *BIT Numerical Mathematics* 66.3 (2026). DOI: 10.1007/s10543-026-01140-w.

Preprints

[38] Y.-T. Liao, H. Luo, and A. Ma. "Efficient and robust Bayesian selection of hyperparameters in dimension reduction for visualization" (2023). URL: https://arxiv.org/abs/2306.00357.

[39] E. H. Bergou, S. Boucherouite, A. Dutta, X. Li, and A. Ma. "Where Have All the Kaczmarz Iterates Gone?" *arXiv preprint arXiv:2510.08563* (2025). DOI: 10.48550/arXiv.2510.08563.

[40] H. Luo, A. Ma, L. Stephan, and Y. Zhu. "Wedge Sampling: Efficient Tensor Completion with Nearly-Linear Sample Complexity". *arXiv preprint arXiv:2602.05869* (2026). DOI: 10.48550/arXiv.2602.05869.

[41] M. F. Serret, A. Cortinovis, Y. Dong, D. Halikias, A. Ma, F. Matti, D. Needell, K. J. Pearce, E. Rebrova, D. Shur, R. Smith, H.-X. Wang, and L. Grigori. "Attention Mechanisms Through the Lens of Numerical Methods: Approximation Methods and Alternative Formulations". *arXiv preprint arXiv:2604.01757* (2026). DOI: 10.48550/arXiv.2604.01757.

Other Works

[42] A. Ma. "Geometry, Intuition, and the Kaczmarz Algorithm". *Girls' Angle Bulletin* 14.6 (Aug. 2021). URL: https://www.girlsangle.org/page/bulletin-archive/GABv14n06E.pdf.

[43] A. Ma. "Getting to the Bottom of It: Gradient Descent and its Variants". *Girls' Angle Bulletin* 15.1 (Oct. 2021). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n01E.pdf.

[44] A. Ma. "When data goes missing". *Girls' Angle Bulletin* 15.2 (Dec. 2021). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n02E.pdf.

[45] A. Ma. “What Is Data: Representing Information as a Matrix”. *Girls' Angle Bulletin* 15.3 (Feb. 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n03E.pdf.

[46] A. Ma. “Data Collection: Challenges and Workarounds”. *Girls' Angle Bulletin* 15.4 (Apr. 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n04E.pdf.

[47] A. Ma. “Data: Classical and Modern Data Visualization Methods”. *Girls' Angle Bulletin* 15.5 (June 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n05E.pdf.

[48] A. Ma. “Supervised Machine Learning: Classification with Candy”. *Girls' Angle Bulletin* 15.6 (Aug. 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv15n06E.pdf.

[49] A. Ma. “The Perceptron Algorithm The Perceptron Algorithm”. *Girls' Angle Bulletin* 16.1 (Oct. 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv16n01E.pdf.

[50] A. Ma. “Natural Language Processing Natural Language Processing”. *Girls' Angle Bulletin* 16.2 (Dec. 2022). URL: https://www.girlsangle.org/page/bulletin-archive/GABv16n02E.pdf.

[51] A. Ma. “Natural Language Processing, Part 2: Language Models”. *Girls' Angle Bulletin* 16.4 (Apr. 2023). URL: https://www.girlsangle.org/page/bulletin-archive/GABv16n04E.pdf.

[52] A. Ma. “Natural Language Processing, Part 3: From Language Models to AI Chatbots”. *Girls' Angle Bulletin* 16.5 (June 2023). URL: https://www.girlsangle.org/page/bulletin-archive/GABv16n05E.pdf.

Professional Activities

Fairness and foundations in machine learning — July 2026, Pasadena, CA
American Institute of Mathematics
Co-organizer

Semester Program: Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications — Spring 2026, Providence, RI
Institute for Computational and Experimental Research in Mathematics (ICERM)
Invited participant for a semester program

Research Collaboration Workshop on Randomized Numerical Linear Algebra — August 2025, Los Angeles, CA
Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: “Randomization in transformer models"

Research Collaboration Workshop in the Science of Data and Mathematics — August 2025, Chapel Hill, NC
University of North Carolina at Chapel Hill
Invited participant for week-long working group
Topic: “Randomized Methods for Tensors"

Collaborate@ICERM — July 2025, Providence, RI
Institute for Computational and Experimental Research in Mathematics (ICERM)
Funded proposal for week-long working group
Topic: “Noise is not Always Bad: A Case Study with the Kaczmarz Algorithm"

Summer Collaborations Program — July 2025, Princeton, NJ
Institute for Advanced Study (IAS)
Funded proposal for two-week-long working group
Topic: “Partial Information Methods for Large-scale Linear Systems"

FOCUS Writing Workshop — April 2025, Berkeley, CA
University of California, Berkeley
Invited to participate in a two-day workshop with grant writing consultant

Mathematics of Intelligences — September 2024, Los Angeles, CA
Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: “Analyzing High-dimensional Traces of Intelligent Behavior"

Research Collaboration Workshop: Women in Data Science and Mathematics — August 2023, Los Angeles, CA

Institute for Pure and Applied Mathematics (IPAM)
Invited participant for week-long working group
Topic: “Graph-based Active Learning"

Women and Mathematics Program May 2022
Los Angeles, CA
Institute for Advanced Study (IAS)
Invited participant for week-long working group
Topic: “The Mathematics of Machine Learning”

Summer Research in Mathematics June 2019
Berkeley, CA
Mathematical Sciences Research Institute (MSRI)
Funded proposal for week-long working group
Topic: “Practical Problems in Distributed Sensing"

Women in Data Science and Mathematics June 2019
Providence, RI
Institute for Computational and Experimental Research in Mathematics (ICERM)
Invited participant for week-long working group
Topic: “Tensor Tools for Multiway Data Analysis"

Visiting Graduate Researcher Spring 2018
Los Angeles, CA
University of California, Los Angeles
Spent one quarter at UCLA as a visiting graduate researcher
Topic: “Sparse and Stochastic Optimization"

Collaborate@ICERM July 2018
Providence, RI
Institute for Computational and Experimental Research in Mathematics (ICERM)
Funded proposal for week-long working group
Topic: “Control and Analysis of Large-Scale Time-Varying Data"

Semester Program: Geometric Functional Analysis and Applications Fall 2017
Berkeley, CA
Mathematical Sciences Research Institute (MSRI)
Invited participant for a semester program

Women in Data Science and Mathematics July 2017
Providence, RI
Institute for Computational and Experimental Research in Mathematics (ICERM)
Invited participant for week-long working group
Topic: “Stochastic Signal Processing for High Dimensional Data"

Gene Golab SIAM Summer School July 2017
Berlin, Germany
Akademie Berlin-Schmöckwitz, Germany
Invited participant for two-week summer school
Topic: “Sparse Recovery and Optimization"

Bayes Impact Spring 2015
San Francisco, CA
Data Science Fellow for 501(c)(3) non-profit Start-up
Worked with an Education Technology company to predict standardized test scores

Graduate Research in Industry Projects (GRIPS) Summer 2014
Berlin, Germany
Institute for Pure and Applied Mathematics (IPAM)
Graduate researcher for inaugural GRIPS program
Topic: Gas Networks with Open Grid Europe

Research in Industry Projects (RIPS) Summer 2013
Los Angeles, CA
Institute for Pure and Applied Mathematics (IPAM)
Undergraduate researcher and project manager for RIPS program
Topic: Linking Social Media to Crime and Disorder

Research Experience for Undergraduate Students Summer 2012
Los Angeles, CA
University of California, Los Angeles
Undergraduate researcher
Topic: Social Networks and Duplicate Detection

Grants, Honors, and Awards

NSF DMS-2606335 09/2026-08/2039

Collaborative Research: Scalable Methods for Incomplete Tensors: To Complete or Not to Complete?
Total award estimate: $91,400

Travel Support for Mathematicians 09/2025-08/2030

Simons Foundation travel support for Mathematicians
Total award estimate: $42,000

Rose Hills Foundation Innovator Grant 07/2024-08/2025

Proposal: "When High- Dimensional Data Goes Missing"
Total award estimate: $50,000

ACHA Charles Chui Young Researcher Best Paper Award 2022

Best Paper Award for "On the number of faces and radii of cells induced by Gaussian spherical tessellations", jointly awarded with Eric Lybrand and Rayan Saab

Rising Stars in Computational and Data Science 04/2019

Rising Stars University of Texas, Austin

UC Chancellor's Postdoctoral Fellowship 2018-2019

Faculty Mentor: Rayan Saab, University of California, San Diego

US Junior Oberwolfach Fellow 10/2018

Mathematisches Forschungsinstitut Oberwolfach

Dissertation Fellowship 2017-2018

Claremont Graduate University

Intellisis Fellowship 2017-2018

Computational Science Research Center at San Diego State University

Intellisis Fellowship 2017

Computational Science Research Center at San Diego State University

Los Alamos National Laboratory Poster Award 2017

Applied Computational Science and Engineering Student Conference (ACSESS)
Computational Science Research Center at San Diego State University

Outstanding Poster Presentation 2013

MAA Undergraduate Poster Session at the Joint Mathematics
Poster: "Linking Social Media to Crime and Disorder", jointly awarded with Y. Hu

Research Mentoring

Ph.D. Advising

Jenny Tran, Graduate Student, UC Irvine 09/2023-current
Emeric Battaglia, Graduate Student, UC Irvine 09/2023-current
Kathryn Dover, Ph.D., Graduate Student, UC Irvine 09/2019-06/2022

- Thesis: Does t-SNE See Curves? Theory and Practice
- First position: Lead Data Scientist at Syniverse
- Co-chaired with Roman Vershynin

Graduate Student Advising

Math 299 - Natanael Alpay, Graduate Student, UC Irvine 03/2025-06/2025

- Topic: Stochastic gradient descent and kernel regression

Math 299 - Nigus Teklehaymanot, Graduate Student, UC Irvine 03/2025-06/2025

- Topic: Tensor-based singular value decomposition

Math 299 - Ashwin Prabu, Graduate Student, UC Irvine 09/2024-03/2025

- Topic: Frontal Slice Descent methods for large-scale tensor systems

Undergraduate Student Advising

Math 199 - Supervised Research Greedy Methods for Linear Feasibility 2024-2025

- James Nguyen (UC Irvine, 2025)
  Adityakrishnan Radhakrishnan (UC Irvine, 2025)
  Oleg Presnykov (UC Irvine, 2026)

Math 199 - Supervised Research Oblique Iterative Projection Methods 2022-2024

- Emi Cervantes (UCI, 2024; First position: PhD in Statistics at UPenn)
  Jenny Tran (UCI, 2024; First position: PhD in Mathematics at UCI)
- Work presented at the Joint Mathematics Meeting 2024 (talk), Symposium for Women and Gender Minorities in Mathematics in Southern California 2024 (talk), Southern California Applied Mathematics Meeting 2024 (poster), and MAA Spring SoCalNev Sectional Meeting 2024 (talk)

Math 199 - Supervised Research Coordinate-wise methods for block missing data 2022-2023

- Meha Patel (UCI, 2023; First position: M.S. in Statistics at CSU Long Beach)
  Sam Rath (UCI 2023; First position: M.S. in Math at San Diego State University)
  Chupeng Zheng (UCI, 2023; First position: M.S. in Data Science at U of Chicago)
- Work presented at Southern California Applied Math Symposium 2023 (poster) and MAA SoCalNev Sectional Meeting 2023 (talk)
- *Published article in SIAM Undergraduate Research Online (SIURO) "Mean Imputation and Stochastic Coordinate Descent for Linear Systems with Missing Data" doi:10.1137/23S1592018*

Math 199 - Supervised Research Stochastic gradient methods for block missing data 2022-2023

- Chelsea Huynh (UCI Irvine, 2023; First position: M.S. in Computer Science at USC) and Michael Strand (UCI, 2023; First position: M.S. in Data Science at UC Irvine)
- Work presented at Southern California Applied Math Symposium 2023 (poster) and MAA SoCalNev Sectional Meeting 2023 (talk)
- *Published proceedings article in IEEE Allerton Conference on Communication, Control, and Computing "Block-missing data in linear systems: An unbiased Stochastic gradient descent approach for Linear Systems with Missing Data" doi:10.1109/Allerton58177.2023.10313429*

IPAM - Research in Industry Projects (Academic Mentor) Traffic Shaping for Real Time Advertising with industry sponsor GumGum Summer 2022

- Aresha Arshad (National University of Sciences and Technology, Pakistan, 2022)
  Gabriela Guerrero Galván (Pomona College, 2023)
  Nico Tripeny (Haverford College, 2023)
  Antonio Plascencia-Vargas (Grand Valley State University, 2023)

IPAM - Research in Industry Projects (Academic Mentor) Online Learning and Matching for Resource Allocation Problems with industry sponsor Alibaba Summer 2019

- Andrea Boscovik (Amherst College, 2021)
  Qinyi Chen (UCLA, 2020)
  Dominik Kufel (University College London, 2021)
  Zijie Zhou (Purdue University, 2021)
- Work presented at Joint Mathematics Meetings 2020 (talk)
- *Published article in SIAM Undergraduate Research Online (SIURO) "Online Learning and Matching for Resource Allocation Problems" doi:10.1137/19S1300534*

UCLA - Research Experience for Undergraduate Students (Graduate Mentor) Topic Modeling and Lyme Disease Data — Summer 2018

UCLA - Research Experience for Undergraduate Students (Graduate Mentor) Matrix Completion and Topic Modeling for Lyme Disease Data — Summer 2017

Teaching

University of California, Irvine Instructor — 2019-current

- Math 13 - Introduction to Abstract Mathematics (W23, SP22)
- Math 105A - Numerical Analysis I (F25, F23)
- Math 105AL - Numerical Analysis I Computing Lab (F25)
- Math 105B - Numerical Analysis II (W24)
- Math 130A - Probability and Stochastic Processes I (SP22, SP21, F20, W20, F19)
- Math 130B - Probability and Stochastic Processes II (W21)
- Math 140A - Introduction to Real Analysis (F21)
- Math 225B - Graduate Numerical Analysis (W25)
- Math 225C - Graduate Numerical Linear Algebra and Special Topics (SP25)
- AISCI 220 - Methods in Machine Learning and AI (FA26)

MSRI - Graduate Summer School Teaching Assistant — Summer 2018

- Representations in High-Dimensional Data

San Diego State University, Department of Mathematics Teaching Assistant — 2016-2018

- Math 151 - Calculus 2: Integration Techniques

Claremont McKenna College, Department of Mathematics Graduate Tutor — 2015-2017

- Math 252 - Statistical Theory, Math 30 - Calculus 1 Differentiation Techniques, CS 55 - Discrete Mathematics and Introduction to Probability
- CCMS Coding lab: Matlab, R, C++, Latex

Professional Activities

UC Irvine STEM Equity Learning Community — 2023-2024

Participated in STEM learning community to discuss equity issues in lower-division STEM courses, particularly Math 13: Introduction to Abstract Mathematics as a part of the Sloan Equity and Inclusion in STEM Introductory Courses program. Presented findings and a call-to-action to the department in Spring 2024.

Noticing in Mathematics for Student Success (NIMSS) Training — 2022-2023

Participated in training provided by UC Irvine Department of Education, which focuses on the study and development of video-annotation curriculum to enhance Instructors' understanding of student thinking in proof-transition courses

Online Remote Teaching Training — 2020

Completed Division of Teaching Excellence & Innovation Online Remote Teaching Training.

Talks and Presentations

Conferences

- Conference on Learning Theory, San Diego, CA — July 2026
  Presented accepted work "Efficient Tensor Completion with Nearly-Linear Sample Complexity"
- The 8th International Conference on Econometrics and Statistics, Tokyo, Japan — August 2025
  Session on Stochastic iterative methods for statistical learning

- The 26th Conf of the International Linear Algebra Society, Kaohsiung, Taiwan — June 2025
  Session on New methods in numerical multilinear algebra
- The 2nd Conf on Random Matrix Theory and Numerical Linear Algebra, Seattle, WA — June 2025
  Conference speaker
- SIAM Conference on Computational Science and Engineering, Fort Worth, TX — March 2025
  Applications of Tensor Methods for Computational and Data Sciences
- Practicum for Undergraduate Mathematicians, Los Angeles, CA — November 2024
  Workshop speaker
- MAA Mathfest, Indianapolis, IN — August 2024
  Iterative and Sketching Approaches for Linear Systems and Beyond
- SIAM Conference on Imaging Science, Atlanta, GA — May 2024
  Recent Challenges and Developments on Multi-way Data Analysis
- SIAM Conference on Applied Linear Algebra, Paris, France — May 2024
  Iterative Methods for Inverse Problems and Uncertainty Quantification
- AWM Research Symposium — October 2023
  Session on Tensor Methods for data modeling
- International Congress on Industrial and Applied Mathematics, Tokyo, Japan — August 2023
  Optimal and Efficient Algorithms for Inverse Problems
- SIAM Conference on Optimization, Seattle, WA — June 2023
  Randomized Methods for Linear Systems: Row-Action and Coordinate Descent
- Center for Approximation and Mathematical Data Analytics Conference, Texas A&M — May 2023
  Conference speaker
- AMS Spring Western Sectional Meeting, Fresno, CA — May 2023
  Special Session on Women in Mathematics
- Pacific Rim Mathematical Association Congress, Vancouver, Canada — December 2022
  Mathematics of Information
- SIAM Mathematics of Data Science, San Diego, CA — September 2022
  Tensor Methods and Applications to Real-World Data
- International Conference on Continuous Optimization, Lehigh University — July 2022
  Optimization for Data Science and Machine Learning
- Women in Inverse Problems Workshop, Virtual — December 2021
  Conference speaker
- SIAM Conference on Applied Linear Algebra, Virtual — May 2021
  Random Matrices and Numerical Linear Algebra
- SIAM Conference on Computational Science and Engineering, Fort Worth, TX — March 2021
  Computational Linear Algebra in Data Analysis
- ICERM Computational Stats and Data-Driven Models Workshop, Providence, CA — April 2020
  Conference speaker
- Women in Mathematics, Southern California, Irvine, CA — March 2020
  Conference speaker
- Joint Mathematics Meeting, Boulder, CO — January 2020
  Special Session on Mathematical and Computational Research in Data Science
- Joint Mathematics Meeting, Boulder, CO — January 2020
  Special Session on The Kaczmarz Algorithm with Applications
- Rising Stars in Computational and Data Science, Austin, TX — April 2019
  Conference speaker
- AWM Research Symposium, Houston, TX — April 2019
  Applied and Computational Harmonic Analysis
- AWM Research Symposium, Houston, TX — April 2019
  WISDM: Women in the Science of Data and Mathematics
- AMS Central and Western Spring Meeting, Honolulu, HI — March 2019
  Special Session on Sparsity, Randomness, and Optimization
- SIAM Conference on Computational Science and Engineering, Spokane, CA — February 2019
  AWM Workshop: Data Science and Mathematics
- Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA — October 2018
  Conference speaker

- Women in Mathematics Southern California Symposium, Los Angeles, CA — February 2017
  Conference speaker

Seminars

- Applied Mathematics Seminar, Michigan State University — April 2026
- RNLA Tutorial, Institute for Computational and Experimental Mathematics — February 2026
- Probability Seminar, Northwestern University — May 2025
- Applied Mathematics Seminar, University of California, San Diego — April 2025
- Applied Mathematics Seminar, University of California, Santa Barbara — February 2025
- Simple Words, Virtual Seminar — November 2024
- Statistics and Data Science Seminar, Rice University — October 2024
- Statistics and Data Science Seminar, Northwestern University — April 2024
- Probability and Combinatorics Seminar, University of California, Irvine — April 2024
- Mathematical Data Science, Purdue-University of Maryland-Penn State Jointly — April 2024
- Probability Seminar, University of Southern California — September 2023
- Data-oriented Math & Statistical Sciences Seminar, Arizona State University — September 2023
- Graduate Seminar, University of California, Irvine — May 2023
- Women and Math Young Researcher Seminar, Institute for Advanced Study — May 2022
- CRM Applied Math Seminar, University of Montreal — March 2022
- Young Data Science Research Seminar, ETH Zurich — February 2022
- One World MINDS Seminar, Virtual Seminar — September 2021
- CMSE Brown Bag Talks, Michigan State University — November 2020
- Graduate Seminar, CSU Channel Islands — May 2020
- Applied Mathematics Seminar, University of California, Merced — September 2019
- Undergraduate Colloquium, University of California, San Diego — March 2019
- Mathematical Sciences Applied Math Seminar, Claremont Colleges Consortium — February 2019
- Center for Computational Mathematics, San Diego State University — December 2018
- Mathematics of Data and Decisions at Davis, University of California, Davis — December 2018
- Summer Colloquium Series, San Diego State University — July 2018
- Level Sets Seminar, University of California, Los Angeles — May 2018
- GFA Young Researchers Seminar, Mathematical Sciences Research Institute — November 2017
- CS Theory Lunch, Stanford University — October 2017
- Algebra, Number Theory, and Combinatorics Seminar, Claremont Consortium — October 2017
- Graduate Seminar, Claremont Graduate University — April 2017

Colloquia

- Mathematical Sciences Mathematics Colloquium, Claremont Colleges Consortium — November 2025
- Department of Mathematics, University of Central Florida — April 2024
- Department of Mathematics, Wellesley College — December 2022
- Mathematics and Statistics Seminar, San Jose State University — September 2017

Posters

- AWM Research Symposium, Minneapolis, MN — June 2022
- Information Theory and Applications, La Jolla, CA — February 2018
- Data Institute Conference at University of San Francisco — October 2017
- Joint Mathematics Meeting, Baltimore, MD — January 2014
- Mathematical Sciences Session, Claremont Consortium, Claremont, CA — September 2014
- AFSOR MURI Fall Meeting, Los Angeles, CA — November 2014
- Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA — November 2014
- Optimization in ML Workshop, NeurIPs, Montreal, Canada — December 2015
- February Fourier Talks, University of Maryland, College Park — February 2016
- ACCESS Meeting, Computational Science Research Center, San Diego, CA — April 2016
- Southern California Applied Math Symposium, Claremont, CA — June 2016

- ACCESS Meeting, Computational Science Research Center, San Diego, CA — April 2017
- Gene Golub SIAM Summer School, Germany — June 2017
- Asilomar Conference on Signals, Systems, and Computers, Monterey, CA — November 2019

Outreach and other presentations

- MathCounts Orange County Chapter Competition, Irvine, CA, *Faculty Lecturer* — February 2025
- UCI Math Circle, Faculty lead — February 2025
- Undergraduate Math Club - Data Science in Math, University of California, Irvine — November 2023
- Computational Science Research Center, San Diego State University, Alumni Panel — April 2023
- UCI Math Circle, Faculty lead — April 2024
- Research in Industrial Projects (RIPS) Celebration 2023, IPAM, Alumni Panel — August 2023
- Math, Computational and Systems Biology Recruitment, UC Irvine, Career Panel — February 2023
- Department of Mathematics Postdocs, UC Irvine, Career Panel — November 2022
- Computational Science Research Center, San Diego State University, Career — February 2019
- Representations of High Dimensional Data, MSRI, Career Panel — July 2018
- The Mathematics of Data Science, MSRI, Graduate Student Panel — July 2018

Service

Conferences and Workshops

- AIMS Workshop "Fairness and Foundations in Machine Learning" — July 2026
- ICERM Workshop, "Randomized Numerical Linear Algebra (RNLA)" — February 2026
- IPAM Research Collaborations, "Randomized Numerical Linear Algebra (RNLA)" — August 2025
- The 26th Conference of the International Linear Algebra Society, Minisymposium — June 2025
- UC Irvine Mathematics of Data, Dynamics, and Life Sciences — March 2025
- Joint Mathematics Meetings, Special Session — January 2025
- SIAM Mathematics of Data Science, Minisymposium — October 2024
- SIAM Linear Algebra and Applications, Minisymposium — May 2024
- Sampling Theory and Applications, Minisymposium — July 2023
- Southern California Applied Math Research Symposium (SoCAMS) — April 2023
- Joint Mathematics Meeting, Special Session — January 2022
- Optimal Transport: Theory, Computation, and Applications Workshop — June 2022
- Southern California Applied Math Research Symposium (SoCAMS) — May 2022
- Women in Math, Southern California (WiMSoCal) — March 2021
- Information Theory and Applications, Minisymposium — February 2020
- AWM UCI Chapter Organizer (Postdoctoral) — 2020-2021
- SIAM Computational Science and Engineering Conference, Minisymposium — February 2019
- University of San Francisco, Data Science Conference, Special Session — February 2019
- Joint Mathematics Meeting, Special Session — January 2019
- CGU Graduate Student Seminar — 2016-2018

Press

- Interviewed for Her Math Story, a platform for showcasing the careers and journeys of modern-day mathematicians, https://hermathsstory.eu/anna-ma/ — January 2025
- Interviewed for article "Greed Works: An Improved Analysis of Sampling Kaczmarz–Motzkin" featured in SIAM Youtube channel, https://www.youtube.com/watch?v=L0zJf_pdpZ0 — June 2024
- Research featured in Claremont Graduate University Alumni Spotlight, https://www.cgu.edu/news/2023/05/anna-ma-phd-in-computational-science-2018-streamlines-the-algorithm/ — May 2023
- Co-organizer for SoCAMS conference featured in SIAM News Blog: "Mathematicians Convene at 2023 Southern California Applied Mathematics Symposium", https://www.siam.org/publications/siam-news/articles/mathematicians-convene-at-2023-southern-california-applied-mathematics — May 2023

-symposium/

Other Activities

- **Coffee/Lunch with a Math Professor Organizer** This program facilitates students interacting more personably with a professor over lunch or coffee. It encourages students and faculty to meet more outside the usual office hours and classroom settings. This program has been an enormous success, reaching over 200 undergraduate students since its inception. 2021-current
- **Cal-Bridge Mentor** The Cal-Bridge program aims to broaden participation by Californians in STEM PhD Programs. As a Cal-bridge mentor, I worked one-on-one with an undergraduate student at a California State University to bridge their time between Cal State and UC and prepare them for applying to PhD Programs. 2024-current
- **Girls' Angle Bulletin Contributing Writer** Girls' Angle is a nonprofit math club founded in 2007. The goal of Girls' Angle is to foster and nurture girls' interest in mathematics and empower them to tackle any field, no matter the level of mathematical sophistication. I was a contributing writer for the bimonthly magazine Girls' Angle Bulletin. 2021-2023

Reviewing and editing

- **Proposals**: NSF, UC Postdoctoral Presidential Fellowship, Simons Foundation
- **Editing:** Co-editor of topical collection on tensor methods in La Matematica, 2024
- **Manuscripts and Conferences**: AAAI 2025 Workshop ColorAI, Advances in Computational Mathematics, Applied and Computational Harmonic Analysis, Applied Mathematics Letters, Applied Numerical Mathematics, BIT Numerical Mathematics, Calcolo, Electronic Transactions on Numerical Analysis, IEEE Letters in Signal Processing, IEEE Transactions on Signal Processing, Inverse Problems and Imaging, Journal of Computational and Applied Mathematics Journal of Machine Learning Research, Journal on Scientific Computing, Linear and Multilinear Algebra, Machine Learning, Numerical Algorithms, Numerical Linear Algebra, Numerical Linear Algebra and Applications, Numerical Mathematics: Theory, Methods and Applications, Numerische Mathematik, Proceedings of the National Academy of Sciences, Sampling Theory and Applications (SampTA), SIAM Journal on Data Science, SIAM Journal on Imaging Sciences, SIAM Mathematics of Data Sciences, SIAM Matrix Analysis and Applications, Signal Processing, Signal Processing Letters, Signal Processing with Adaptive Sparse Structured Representations (SPARS)